Multi-Agent Passion Intelligence

Hire the people who can't stop building what you're building

A swarm of Gemma agents on an AMD Instinct MI300X reads a builder's public code, hackathons and writing — and finds the engineers who never applied.

`Gemma 4 31B` `AMD Instinct MI300X` `ROCm` `Ollama` `LangGraph` `pgvector · Cloud SQL` `FastAPI · Next.js` `Google Cloud TTS`

AMD Developer Hackathon · ACT II — Track 3 🦄 Unicorn · passion-intelligence.vercel.app

The Problem

Hiring reads the one document written to be read — and misses the record nobody writes for a recruiter.

- **A resume is a sales document.** It says what someone *wants* you to believe.
- **The best engineers never apply.** They already have jobs. They don't know your startup exists.
- **Keyword screening is theatre.** "5 years of Python" tells you nothing about whether they care.
- **Startups can't outbid Big Tech** on salary, equity story, or brand.
- **Sourcing is expensive.** A recruiter seat and a contract sourcer are out of reach at seed stage.

~70%

of the workforce is **passive** — not looking, and therefore invisible to every job board

~30%

of first-year salary — the commonly cited cost of a **bad hire**

Industry estimates — verify before publishing

Why AI is Essential

You cannot rule-base this. Judging whether someone is genuinely obsessed with a problem means reading what they built — code, prose, and pictures — and forming a judgement.

Passion is inferred, never declared

Nobody writes "I am obsessed with wildfire detection" on a resume. It has to be **inferred** from a pattern of voluntary work across years. That is a reasoning task, not a query.

The evidence is multimodal

Half the signal is in **pictures** — the architecture diagram, the dashboard screenshot, the model output. Gemma's vision model reads them and reports what the work demonstrates.

Fit is semantic, not keyword

"CRDT" and "conflict-free replication" are the same thing. Embeddings on the MI300X feed a **pgvector** search, so we match meaning instead of strings.

Search must start from a mission

Free Discovery takes a **project description** and turns it into a search for humans. No keyword list, no boolean string, no candidate names.

Why This Wins for Startups

Track 3 asks for product/market potential. Here is the wedge: startups lose the hiring game on every axis except one — and that axis is exactly what we measure.

They can't win on money

Big Tech will always pay more, vest faster, and de-risk the offer. Competing there is a losing trade.

They can win on mission fit

Somewhere there are five engineers already building your exact thing on weekends. They'd take a pay cut to do it full-time.

We find those five

Not by screening applicants. By reading the open internet for people whose **voluntary work** already points at your mission.

The buyer is obvious. A seed-stage founder making their first five engineering hires, with no recruiter, no sourcing budget, and no brand. Today they post to a job board and pray. We hand them a ranked, contactable shortlist of people who were *already* building their product before they existed.

What Hiring Tools Don't Do Today

ATS platforms and recruiter seats are built to *filter people who applied*. Nothing on the market starts from your mission and goes looking.

They only see applicants

An ATS ranks the pile you already have. If the right person never applied — and they didn't — the tool is structurally blind to them.

They read the resume, not the work

Nobody opens the repository. Nobody looks at the architecture diagram. The actual evidence of skill goes unread.

They match keywords, not meaning

Boolean strings on a profile. A person can be perfect for you and share zero keywords with your job description.

They can't tell you *why*

A score with no citation is a vibe. There is no link to click, no repo to read, nothing to check.

Our Solution: Passion Intelligence

Give it a mission. Ten Gemma agents investigate the open internet and come back with people, evidence, and a pitch.

🔭 Free Discovery

Supply **no names at all**. It searches GitHub from your mission and finds builders who never applied.

📋 Applicants

Only a **GitHub handle** is required. It auto-discovers their blog, Dev.to, Kaggle, Devpost, **lablab.ai** and hackathons.

👁️ Multimodal evidence

Gemma **vision** reads architecture diagrams and product screenshots. Photos of people are filtered out.

🔗 Cited scores

Every number links to a **real URL**. Click through and read the repo yourself. Nothing is invented.

📍 Contactable only

A candidate is only **selected** if there's a LinkedIn to reach them on. A map shows where they are.

🎬 A pitch you can forward

A narrated video, captioned **twice** — once for a technical hiring manager, once for HR.

Two Ways In: Source or Screen

The same engine, pointed in two directions — and the first one is the part nobody else does.

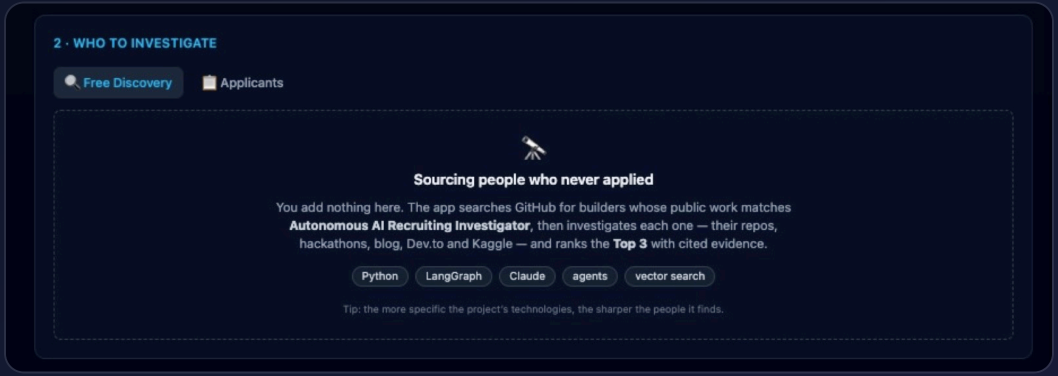
🔭 Free Discovery — sourcing

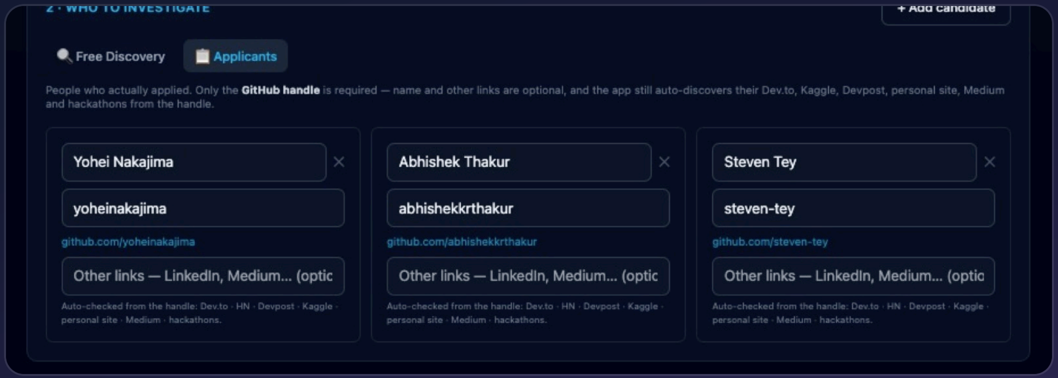
You provide **nothing** about people. Just the mission. It searches GitHub, ranks by fit, and investigates strangers who never applied and never would have found you. **This is the original idea.**

📋 Applicants — screening

For the pile you already have. Only the **GitHub handle** is required — name and links are optional. Dev.to, Kaggle, Devpost, **lablab.ai**, Medium and their personal site are auto-discovered.

`no names required` `handle only` `evidence-cited`





How It **Works**

From a project description to a contactable, evidence-backed shortlist.

STEP 1
Describe the mission
What you're building, and the tech it needs

→

STEP 2
Find the humans
GitHub search from the mission alone

→

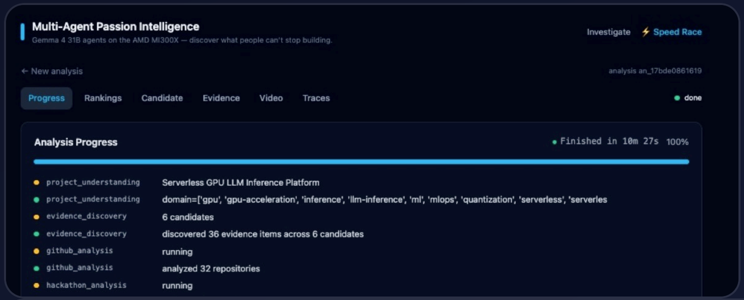
STEP 3
Investigate
Repos, hackathons, Kaggle, writing, images

→

STEP 4
Rank + cite
Weighted score, every claim linked

→

STEP 5
Pitch it
Narrated video for two audiences



The 10-Agent Pipeline

Orchestrated with LangGraph. Every agent writes into shared state; every LLM call runs on the MI300X.

1 · Project Understanding
Turns the mission into domains, features and technologies — the fingerprint everything else is matched against.

2 · Evidence Discovery
GitHub, Dev.to, Hacker News, Devpost, Kaggle, personal sites. Every item keeps its source URL.

3 · GitHub Analysis
Repo quality, architecture, maturity, what they actually ship.

4 · Hackathon Analysis
What they built when there was no salary attached — the purest passion signal there is.

5 · Visual Portfolio `vision`
Gemma **sees** the diagrams and screenshots and judges design, rigor and domain.

6 · Passion Detection
Genuine passion, consistency, innovation, voluntary effort.

7 · Similarity `pgvector`
Tag overlap + embeddings from the same GPU, searched in Cloud SQL.

8 · Ranking
Transparent weighted score. LinkedIn required to be selected.

9 · Storytelling
The human explanation — every sentence citing an evidence id.

10 · Recommendation Video
Rendered, narrated, and captioned for two audiences.

Evidence, Not Vibes

An AI that scores people has to be auditable, or it is worse than useless. This one fails its own build if it invents anything.

Every claim carries a URL

If the system says someone is obsessed with wildfire detection, you click through and read the repository. There is nothing to take on faith.

The test suite enforces it

18 tests. A score citing evidence that doesn't exist, or a narrative referencing an invented project, **fails the build**.

Scores are deterministic

Gemma writes the prose and reads the images. The *numbers* come from transparent, reproducible weights — not from a model's mood.

```
$ pytest -q

  test_every_score_has_evidence             PASSED
  test_narrative_projects_are_real          PASSED
  test_no_evidence_without_source_url       PASSED
  test_similarity_top_evidence_are_real     PASSED
  ...

  18 passed

# every ranked candidate must cite real evidence ids
# every narrative project must map to a real URL
# no evidence may exist without a source
```

Built on AMD — three modalities, one GPU

Not a hosted API. Reasoning, vision *and* embeddings all execute on the same AMD Instinct MI300X through ROCm.

🧠 Reasoning — Gemma 4 31B

Open weights, 23 GB, **100% resident on the GPU**. Understanding, passion, ranking, storytelling.

👁️ Vision — the same model

The CLIP encoder runs on the **ROCm backend**. It reads architecture diagrams and screenshots.

🧮 Embeddings — all-minilm

384-dim vectors from the **same MI300X**, feeding pgvector similarity search.

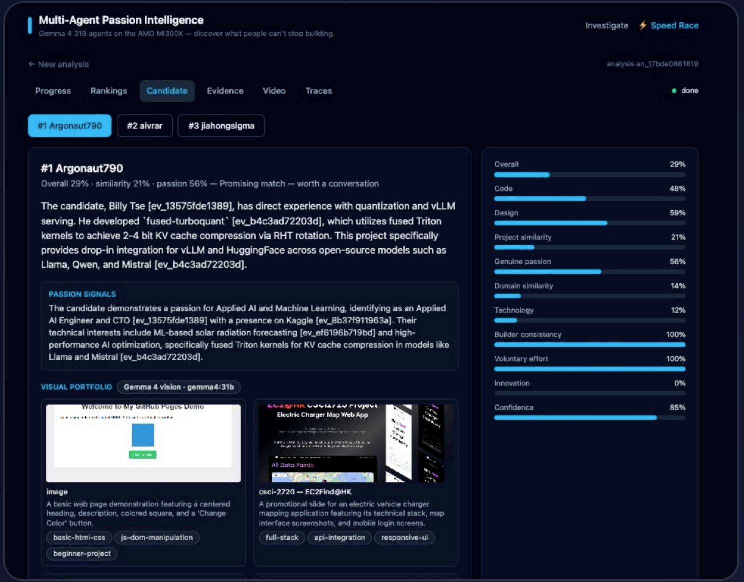
`AMD Developer Cloud` `ROCm` `ollama/ollama:rocm`

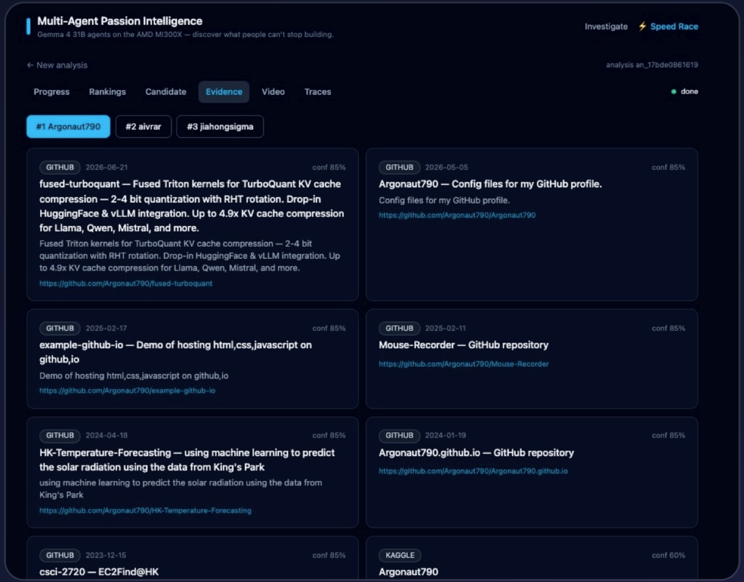
`100% open-source models`

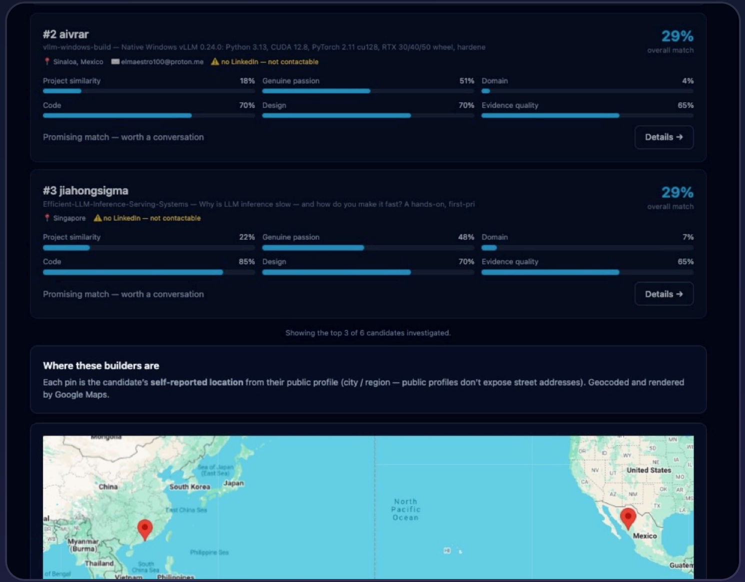
```
$ rocm-smi
  Card Series: AMD Instinct MI300X
  Power: 274 W   Temp: 43 °C   VRAM: 205 GB

$ ollama ps
  NAME          SIZE    PROCESSOR    CONTEXT
  gemma4:31b    23 GB   100% GPU     262144

$ docker logs ollama | grep ROCm
  clip_ctx: CLIP using ROCm0 backend
  sched_reserve: ROCm0 compute buffer 1752 MiB

# text + vision + embeddings — one AMD GPU
```

Seen. Cited. Reachable.

The three things a hiring tool has to earn — and the three nobody else does. All from a single Free Discovery run, nobody supplied.







Gemma sees the work

Vision reads their architecture diagrams and product screenshots — **gemma4:31b**, on the MI300X — and says what the work demonstrates.

Every claim has a URL

The narrative cites evidence ids; the explorer shows the source behind each one. **Click it and read the repo yourself.**

People you can reach

Self-reported location, geocoded to city/state/country. **No LinkedIn → flagged, not selected.** A ranking you can't act on is a list of nobody.

System **Architecture**

The backend runs *on* the GPU host, so Gemma is reached over localhost — no tunnel, nothing to fall over.

Client

Next.js dashboard on Vercel

Live SSE progress · Rankings + map · Evidence explorer · Video with synced captions

Application — AMD droplet

FastAPI (systemd) behind **Caddy** TLS

LangGraph 10-agent pipeline · ffmpeg + Pillow video render · Google Cloud TTS narration

Intelligence & state

Ollama / ROCm on the MI300X — localhost:11434

Gemma 4 31B (text + vision) · all-minilm embeddings · **Cloud SQL + pgvector** · Langfuse traces

Technology **Deep Dive**

Why each piece is here — and what we deliberately refused to claim.

The stack that runs

Gemma 4 31B `open weights`

One model for reasoning **and** vision. No proprietary model on the critical path.

Ollama + ROCm on MI300X

Serving layer. Container `ollama/ollama:rocm`, bound to `/dev/kfd` and `/dev/dri`.

LangGraph · FastAPI · Next.js

Ten agents over shared state, streamed to the browser over SSE.

Cloud SQL + pgvector · Google Cloud TTS

Semantic search and persistence; a Studio-voice narrator for the video.

Bright Data `Web Unlocker`

Unblocks LinkedIn, Medium and lablab.ai so the bot-protected half of a candidate's work is readable.

Reaching what the web hides

Bright Data — Web Unlocker

LinkedIn, Medium and lablab.ai are bot-blocked to a plain scraper. **Bright Data unblocks them**, so the contact trail and the writing get read like any other source.

Everything else, read directly

GitHub, Dev.to, Hacker News, Devpost, Kaggle and personal sites need no unblocking — and **every item keeps its source URL**.

Why it matters

A candidate you cannot contact is a dead end. Unblocking the profile layer is what turns a ranked list into **people you can actually reach**.

Every model in the stack is open source — swap the GPU host and the whole system still runs.

Everyone Wins

The rare hiring product where the candidate isn't the one being squeezed.

For the startup

- ✓ **Reach passive talent** without a recruiter seat
- ✓ **Compete on mission**, not on salary
- ✓ **Shortlist in minutes**, not weeks of sourcing
- ✓ **Every claim auditable** — click the repo
- ✓ **No per-seat SaaS tax** — open models

For the hiring manager

- ✓ **Read the work**, not the resume
- ✓ **See the diagrams** they actually drew
- ✓ **A ranked case** with cited evidence
- ✓ **A video** to forward to your co-founder
- ✓ **Contactable only** — no dead ends

For the candidate

- ✓ **Judged on what you built**, not how you write CVs
- ✓ **Found for the work you love** — the weekend project
- ✓ **No application** to fill in. Ever.
- ✓ **Public work only** — nothing private is touched
- ✓ **Offered the job** you'd have done anyway

Market & Business Model

Track 3 asks for product/market potential. Sourcing is the expensive half of hiring — and the half nobody has automated.

Who pays

Seed → Series B startups making their first 5–50 engineering hires. No recruiting team. No brand. Highest pain, fastest decision, one buyer: the founder.

Why they pay

A contract sourcer or a recruiter seat costs **thousands per year** and still only searches profiles. One good hire pays for the product many times over.

Why now

Open models are finally good enough to **read code and images** and reason about them — and cheap enough to run on your own GPU.

Wedge

Free Discovery for a single role. Land with the thing nobody else does — sourcing people who never applied.

Expand

Per-role pricing → team seats → the talent graph: everyone on the open internet building toward your domain.

Moat

The evidence graph compounds. Every analysis deepens the map of who builds what — and cited evidence is defensible where vibes are not.

It's Live — go and break it

Not a notebook. A deployed product: hosted API, real database, persisted analyses, and a test suite that fails the build on a fabricated citation.

Try it

passion-intelligence.vercel.app

API · 129-212-179-131.sslip.io/api/health

What to do

Run the seeded demo → watch ten agents work live → open **Rankings** for cited scores, contact details and the map → open **Video**, press play, and switch the captions between **tech** and **HR** mid-playback.

`18/18 tests` `10 agents` `Gemma on MI300X` `evidence-cited`

Resumes tell you what people want you to believe. We read what they build when nobody is watching.